

DemandSense

Private Label Demand Intelligence · Irving Tissue Toronto Plant

The Problem

Irving Tissue converts tissue rolls into finished private-label SKUs for Loblaws, Walmart, Costco, Metro and Sobeys at its Toronto facility. Each retailer has distinct demand patterns driven by promotions, seasonal flu peaks, and holiday stockpiling. When forecasts are wrong, the plant either overproduces (waste, cost) or underproduces (missed fill rates, retailer penalties). With a \$600M Macon expansion scaling output by 50%, the cost of poor forecasting compounds across a larger network.

What DemandSense Does

■ Demand Forecasting	Prophet + XGBoost models at the SKU × retailer level. 12-week horizon with 90% confidence intervals. Promotional spikes modelled as regressors. MAPE: ~13% on holdout.
■ Anomaly Detection	Isolation Forest flags demand surges and supply shocks before they become missed orders — giving the plant early warning to re-sequence production runs.
■ Sustainability Quantifier	Connects forecast accuracy to Irving's 2030 commitments. A 10% MAPE improvement prevents 3,300+ waste units/yr and ~397 kg of tissue from landfill — measurable progress toward the 90% diversion target and ISO 14068-1:2023 carbon-neutral supply chain.

Model Results (Synthetic Data — Realistic Simulation)

Metric	Value	vs. Industry Benchmark
Prophet MAPE (holdout)	~13%	Benchmark: 15–25% (CPG private label)
Anomaly recall	~78%	Isolation Forest, 5% contamination rate
Waste units prevented	3,308/yr	At 10% MAPE improvement
Retailer × SKU combos	20	5 retailers · 4 SKUs modelled
Forecast horizon	12 weeks	Weekly cadence, rolling retrain

Tech Stack

Python · Facebook Prophet · scikit-learn (Isolation Forest) · SQL · Streamlit · Plotly · Azure-ready pipeline architecture · Power BI compatible outputs

Built by a Software Engineering Technology — Artificial Intelligence student · Data Modelling Co-op application · Irving Tissue Toronto · Fall 2026 | [github.com/\[your-username\]/DemandSense-Irving](#)